

Hypothesis Stress-Test & Investment Decision

Report v3.0.6.8

Powered by Vraimony

Protocol: ImmuneErrorRadar falsification protocol v3.0.6.9.7 | method=Vraimony methodology
| posture=standard_followed_not_copied

Falsification receipt: id=ierfp-85503814e099ba54d16f0f7a |
contentSha256=85503814e099ba54d16f0f7a | signedByVraimony=false

Documentation standard: ERF-inspired evidence receipt discipline: hash-first, portable,
replayable, claim-bounded.

Citation line: This report follows the ImmuneErrorRadar falsification protocol, powered by
Vraimony, based on the Vraimony evidence-receipt methodology.

Vraimony boundary: This receipt documents a falsification workflow and report payload
hash; it is not medical advice, not clinical validation, not wet-lab evidence, and not
proof of biology.

Report mode: verification-only | evidence-readiness receipt | investment-triage verdict |
no discovery claim | no clinical advice

FINAL INVESTMENT DECISION v3.0.6.7: DO_NOT_INVEST_CLAIM_KILLED | claim=claim killed as
written for current data | killedClaim=bulk_surrogate_is_true_gamma_delta_T_cell_evidence
| criteria=kill:6/invest:0 | next=Do not invest in gamma-delta claim from bulk data.

Acquire TRGC/TRDV/TCR/single-cell evidence first. | probabilityReported=false

PREPRINT READINESS v3.0.6.8: PREPRINT_READY_CLAIM_KILLED_AS_WRITTEN | title=Hypothesis
Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true |

residualDataFollowUp=false | wetLabAllowed=false | probabilityReported=false

Publication data fuel v3.0.6.9: route=GAMMA_DELTA_NKG2D_SURROGATE |

state=NO_ROUTE_SPECIFIC_PUBLICATION_FUEL_BEYOND_EXISTING_KILL_INVEST_DECISION |

syntheticData=false | fakeReplication=false | defense=No additional route-specific data

fuel changes the final decision; publish only criteria-based triage. | APM multi-cohort

fuel v3.0.6.9.6: state=APM_MULTI_COHORT_DATA_FUEL_INJECTED_VALIDATION_TARGETS_ONLY |

syntheticData=false | fakeReplication=false | targets=medulloblastoma, Wilms

tumor/nephroblastoma, Ewing sarcoma | boundary=data-fuel only, not replication.

Pediatric cross-tumor scope v3.0.6.9.6: state=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE |

crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false

Required external pediatric tumors: medulloblastoma, Wilms tumor / nephroblastoma, Ewing
sarcoma

Investment triage v3.0.6.6: verdict=KILL_TRUE_GAMMA_DELTA_CLAIM_SURROGATE_ONLY |

killedClaim=bulk_surrogate_is_true_gamma_delta_T_cell_evidence | basis=kill criteria

dominate and no defensible next investment exists for the supplied claim; kill=6, invest=0

| next=Do not invest in gamma-delta claim from bulk data. Acquire

TRGC/TRDV/TCR/single-cell evidence first. | probabilityReported=false

Question understood: true

Question: In pediatric neuroblastoma GSE85047, high expression of $\gamma\delta$ T cell marker genes

(TRGC1, TRDV1, TRDV2) reflects greater $\gamma\delta$ T cell infiltration and predicts better

progression-free survival independently of MYCN amplification. This survival advantage is

mediated through NKG2D-MICA/B recognition rather than classical MHC-I-dependent

cytotoxicity, making it independent of B2M/HLA expression status.

Question understanding v3.0.3.97: GAMMA_DELTA_T_CELL_NKG2D_SURVEILLANCE | confidence=0.95

Mechanistic text understanding v3.0.4.3: GAMMA_DELTA_T_CELL_NKG2D_SURVEILLANCE_TEXT |

confidence=0.96

Text source context: gamma-delta T cells in pediatric neuroblastoma style text |

prior=EMERGING_PEDIATRIC_NEUROBLASTOMA_GAMMA_DELTA_TIL_AXIS

Text genes understood: B2M,TRGC1,TRDV1,TRDV2,MICA,TRGC2,TRDC,KLRK1,MICB,NKG7,GZMB

Mechanism claim parsed: $\gamma\delta$ T cells may recognize neuroblastoma stress ligands through NKG2D/MICA/B and provide MHC-independent immune surveillance.

Local proxy tests:

GAMMA_DELTA_SURVEILLANCE_PROXY:PROVISIONAL_SURROGATE_EXECUTABLE_FULL_TRGC_TRDV_MAPPING

Not directly testable here: $\gamma\delta$ TCR clonotype/function,autologous tumor killing,NKG2D

blocking experiment,TIL product composition

Why this route: $\gamma\delta$,TRGC1,TRDV1,TRDV2,NKG2D,MICA

Cancer genes understood: MICA,KLRK1,B2M,HLA-A,MYCN,TRGC1,TRDV1,TRDV2,TRGC2,TRDC,MICB

Dataset/outcome resolved: GSE85047 | PFS | covariates=MYCN

Literature axis basis: $\gamma\delta$ T-cell/NKG2D text is decomposed into immune-cell infiltration markers plus NKG2D-MICA/B context; functional recognition is not proven by bulk expression.

Understanding caveats: gamma_delta_t_cell_markers_may_be_missing_from_selected_matrix

Hypothesis tested: GAMMA_DELTA_NKG2D_MICA_SURVEILLANCE_PFS_MYCN_GSE85047

Primary axis: MICA_NK_DECOUPLING |

route=PRIMARY_AXIS_GAMMA_DELTA_NKG2D_SURVEILLANCE_TEXT_PROXY

Genes understood: MICA,KLRK1,B2M,HLA-A,MYCN,TRGC1,TRDV1,TRDV2,TRGC2,TRDC,MICB

Dataset/model: GSE85047 | outcome=PFS | model=gamma-delta T-cell/NKG2D text proxy with selected-matrix surrogate — TRGC/TRDV targets injected as provisional surrogate, not verified full GPL5175 probes

Initial result: INITIAL_EXPLORATORY_ADVERSE_SIGNAL_REVIEW_ONLY |

priority=SECONDARY_LEAD_REVIEW_ONLY

Triage signal: REVIEW_ONLY_DIRECTIONAL_SCREEN |

level=REVIEW_ONLY_DIRECTIONAL_SCREEN_CANDIDATE |

verdict=REVIEW_ONLY_COMPUTATIONAL_ROUND_OFFICIAL_CLAIM_BLOCKED

Signal promotion: rawAllowed=true | forKilledClaimAllowed=false |

residualDataFollowUpAllowed=false | nextComputationalRoundCandidate=true |

wetLabCandidate=false | officialClaimAllowed=false

Directional decision v3.0.4.1:

GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED |

local=KEEP_AS_SURROGATE_DIRECTION_PENDING_VERIFIED_TRGC_TRDV_MAPPING | rank=6

Best direction scan: APM_COMPOSITE_PFS_MYCN_GSE85047 | STRONG_EXPLORATORY_DIRECTION | score=12.3647

Directional ranking table: #1 APM composite:STRONG_EXPLORATORY_DIRECTION | #2

B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION | #3 CK2/APM

fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #4 HDAC/APM

fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #5 NK

effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE | #6

$\gamma\delta$ /NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED | #7

CD276/B7-H3:SUBGROUP_ONLY_DIRECTION

Lead label: GAMMA_DELTA_SURROGATE_DIRECTION_VERIFY_TRGC_TRDV_BEFORE_CLAIM

Directional why: A $\gamma\delta$ /NKG2D surrogate direction was executed using injected

selected-matrix surrogate probes; this is not verified TRGC/TRDV expression and requires full GPL5175 mapping: delta=0.161765, adjustedHR=1.2228/p=0.207.

Directional next action: Acquire official TRGC/TRDV/TRDC GPL5175 mappings and, if possible, TCR/single-cell/TIL functional data before treating the surrogate as $\gamma\delta$

evidence.

Proxy attribution v3.0.4.4: GAMMA_DELTA_SURROGATE_EXECUTED_NOT_TRGC_TRDV_VERIFIED |
signalSource=GAMMA_DELTA_SURROGATE_FROM_SELECTED_CYTOTOXIC_NKG2D_CONTEXT |
mechanismGeneTested=false |

label=GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED

Probe lookup v3.0.5.0: resolvedVector=8 | resolvedButVectorMissing=20 | probeUnknown=0 |

absentOrReporterUnavailable=2 |

staticManifestLocalResolve=TARGETED_STATIC_MANIFEST_ATTACHED | runtimeNetwork=false |

legacyAutoFetchFlag=true

Probe vector binding v3.0.5.1: columnIndexPresent=true | resolvedProbelds=28 |

boundVectors=8 | resolvedButColumnMissing=20 | aliasBoundVectors=0 |

nextBlocker=PARTIAL_BINDING_REMAINING_GENES_NEED_COLUMN_EXTRACTION

Evidence fuel v3.0.6.0: GAMMA_DELTA_NKG2D_SURROGATE |

GAMMA_DELTA_NKG2D_SURROGATE_FUEL_AVAILABLE_REVIEW_ONLY |

expected=GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED |

boundary=DATA_FUEL_IS_TEST_RECIPE_AND_SELECTED_RUNTIME_COVERAGE_NOT_TIER2_PROOF

$\gamma\delta$ probe injection v3.0.4.4: state=PROVISIONAL_GAMMA_DELTA_SURROGATE_READY_REVIEW_ONLY |

exactTRGC/TRDV=false | usableRelaxed=true

Executed local directions v3.0.4.4: #1 APM

composite:STRONG_EXPLORATORY_DIRECTION[APM_COMPOSITE_DIRECT_PROXY] | #2

B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION[B2M_TAP1_NK_DIRECT_PROXY] | #5 NK

effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY]

| #6

$\gamma\delta$ /NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED[GAMMA_DELTA_S

| #7 CD276/B7-H3:SUBGROUP_ONLY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY] |

#8

ODC1/polyamine:WEAK_EXPLORATORY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY]

Mechanistic hypotheses needing assets v3.0.4.4: CK2/APM

fallback:APM_FALLBACK_STRONG_SIGNAL_CK2_MECHANISM_NOT_DIRECTLY_TESTED | HDAC/APM

fallback:APM_FALLBACK_STRONG_SIGNAL_EPIGENETIC_MECHANISM_NOT_DIRECTLY_TESTED |

$\gamma\delta$ /NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED |

NECTIN2/TIGIT:NECTIN2_TIGIT_NOT_TESTABLE_PROBE_AND_SINGLE_CELL_CONTEXT_REQUIRED

N=272 | events=96 | highEventRate=0.433824 | lowEventRate=0.272059 | delta=0.161765

Cox first-pass: state=RELAXED_COX_FIRST_PASS_EXECUTED_REVIEW_ONLY | HR=1.4279 | p=0.0098 |

rows=272 | events=96

MYCN-adjusted Cox first-pass: state=RELAXED_COX_FIRST_PASS_EXECUTED_REVIEW_ONLY |

HR=1.2228 | p=0.207 | rows=272 | events=96

MYCN-stratified event-rate: nonMYCN delta=0.072575 n=218 | MYCN-amp delta=0.107143 n=54

Component separation: not applicable

Tier-1 package v3.0.4.2: TIER1_FULL_GPL5175_LOCAL_COX_TARGET |

current=TIER0_DIRECTIONAL_SELECTED_MATRIX_ACTIVE | noStrictnessIncrease=true

GPL5175 annotation: FULL_GPL5175_ANNOTATION_REQUIRED_BEFORE_TIER1_PASS | fullPass=false

Verified-local Cox package: VERIFIED_LOCAL_COX_RERUN_CANDIDATE_RELAXED_MODE |

label=SECONDARY_AXIS_VERIFIED_LOCAL_COX_RERUN_TARGET_GAMMA_DELTA_SURROGATE_DIRECTION_TRG

Reproducibility receipt:

sha256-c198165194c3be812b1f5f2d5d6c57dcb872f9a0e1eff1da7d6b9ad268494c53 |

replay=REPLAY_PACKAGE_SPEC_READY_CURRENT_RUN_EMBEDDED_JSON_ONLY

Method claim: The app parses cancer hypotheses, maps them to expression-survival axes,

runs relaxed first-pass event-rate/Cox screens, ranks directional signals, and packages

reproducibility receipts while blocking official proof/kill claims before Tier-2.

Biology claim: This axis is interpreted only as a local directional result in the current selected-matrix screen.

Evidence readiness gate: BLOCKED_BY_MISSING_EVIDENCE |

missing=official_or_declared_probe_platform_mapping,probe_to_gene_mapping_receipt,independent_replication_o

Evidence readiness receipts: platform/probe=PARTIAL_MAPPING_REVIEW_ONLY |

outcomeResolved=true | covariateResolved=true |

phenotypeComparator=SURVIVAL_OUTCOME_MODEL_NO_PHENOTYPE_COMPARATOR_REQUIRED_OR_NOT_DEC

| replicationComparator=REPLICATION_COMPARATOR_NOT_DECLARED |

confounder=CONFOUNDER_ADJUSTED_MODEL_EXECUTED_REVIEW_ONLY | mycnCoxExecuted=true |

fragility=RAW_SIGNAL_STRONG_BUT_WEAKENED_AFTER_MYCN_ADJUSTMENT_REVIEW_ONLY

Axis continuation: NO_AXIS_CONTINUATION_FROM_CURRENT_VERDICT | next=Keep current

evidence-readiness verdict; no axis continuation is opened.

Operational readiness: BLOCKED_FROM_PROMOTION_BY_READINESS_GAPS — Resolve first:

official_or_declared_probe_platform_mapping, probe_to_gene_mapping_receipt,

independent_replication_or_cohort_transfer_check

Final hypothesis state v3.0.6.0:

GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED | reason=Gamma-delta/NKG2

is a distinct surrogate route; it must not inherit B2M/APM or CD276 final states. Current

local execution is provisional and requires official TRGC/TRDV/TRDC mapping before any

gamma-delta evidence language.

Decisive adjudication v3.0.6.5:

verdict=NOT_TESTABLE_AS_GAMMA_DELTA_EVIDENCE_SURROGATE_ONLY | claimKilled=false |

localSignalSurvives=false | notTestable=true | action=Do not run pseudo-Cox or proxy

promotion; acquire the missing platform/single-cell/functional data first.

Investment verdict v3.0.6.6: verdict=KILL_TRUE_GAMMA_DELTA_CLAIM_SURROGATE_ONLY |

killedClaim=bulk_surrogate_is_true_gamma_delta_T_cell_evidence | killCriteria=6 |

investCriteria=0 | next=Do not invest in gamma-delta claim from bulk data. Acquire

TRGC/TRDV/TCR/single-cell evidence first.

FINAL INVESTMENT DECISION v3.0.6.7: DO_NOT_INVEST_CLAIM_KILLED | claim=claim killed as

written for current data | killedClaim=bulk_surrogate_is_true_gamma_delta_T_cell_evidence

| criteria=kill:6/invest:0 | next=Do not invest in gamma-delta claim from bulk data.

Acquire TRGC/TRDV/TCR/single-cell evidence first. | probabilityReported=false

PREPRINT READINESS v3.0.6.8: PREPRINT_READY_CLAIM_KILLED_AS_WRITTEN | title=Hypothesis

Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true |

residualDataFollowUp=false | wetLabAllowed=false | probabilityReported=false

Publication data fuel v3.0.6.9: route=GAMMA_DELTA_NKG2D_SURROGATE |

state=NO_ROUTE_SPECIFIC_PUBLICATION_FUEL_BEYOND_EXISTING_KILL_INVEST_DECISION |

syntheticData=false | fakeReplication=false | defense=No additional route-specific data

fuel changes the final decision; publish only criteria-based triage. | APM multi-cohort

fuel v3.0.6.9.6: state=APM_MULTI_COHORT_DATA_FUEL_INJECTED_VALIDATION_TARGETS_ONLY |

syntheticData=false | fakeReplication=false | targets=medulloblastoma, Wilms

tumor/nephroblastoma, Ewing sarcoma | boundary=data-fuel only, not replication.

Pediatric cross-tumor scope v3.0.6.9.6: current=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE |

observed=neuroblastoma | crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false

Pediatric expansion targets: medulloblastoma,Wilms tumor / nephroblastoma,Ewing sarcoma

Pediatric scope next action: After neuroblastoma, run the same stress-test template on

medulloblastoma, Wilms tumor/nephroblastoma, and Ewing sarcoma; keep each as KILLED /

NULL-LIKE / SURVIVING / NOT_MEASURABLE with separate receipts before any cross-tumor claim.

Pediatric scope boundary: Neuroblastoma-only evidence remains single-cohort and

disease-scoped. Cross-pediatric-tumor credibility requires independent human datasets from at least three additional pediatric tumor families before any generalization, Tier-2 promotion, or wet-lab language.

Preprint boundary: This report supports a methods/resource-allocation claim only: supplied computational hypotheses are triaged as killed-as-written, watchlist-only, or data-only follow-up. It does not prove or disprove biology, does not recommend treatment, and does not establish clinical actionability.

Allowed manuscript sentence: The supplied claim is killed as written for the current data; the result should redirect resources away from the claimed mechanism rather than imply biological impossibility.

Direction: HIGH_SCORE_WORSE_PFS_EVENT_RATE | caveat=Relaxed first-pass screen only. Shows trial event-rate, MYCN-stratified and in-app Cox-review numbers for debugging. Not proof, not official support/kill, not clinical, not publication-grade, and not Tier-2 computational evidence.

Missing gates: official TRGC1/TRGC2/TRDV1/TRDV2/TRDC full GPL5175 mapping still required,current execution uses provisional selected-matrix $\gamma\delta$ /NKG2D surrogate,NKG2D blocking/autologous killing not available,TIL product composition not available,full GPL5175 audit,source certification,independent Cox-capable replication,Tier-2 not reached,full GPL5175 mapping required before official support/kill,independent replication required before computational evidence language

Claim guard: BLOCKED_COMPUTATIONAL_CLAIM_REQUIRES_TIER2 | tier=TIER0_SLICE_REPRODUCIBLE_REVIEW_ONLY | officialPromotionAllowed=false | experimentalPromotionAllowed=false | experimentalPromotionForKilledClaimAllowed=false | dataFollowUpAllowed=false | computationalReplicationAllowed=false | wetLabAllowed=false | officialClaimAllowed=false

Tier-0 proves replayability, not source truth.

Tier-1 proves source extraction, not biological truth.

Tier-2 is required before any computational evidence claim.